

Circulant Ramsey(5,5) Colorings: Maximum Order 41 and the Exclusion Band 42–46

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Abstract

We determine, by exhaustive computation, the complete spectrum of *circulant* Ramsey(5,5) colorings on n vertices for every $5 \leq n \leq 46$. A circulant Ramsey(5,5) coloring is a red/blue edge-coloring of K_n in which both color classes are circulant (vertex-transitive under $\mathbb{Z}_n$) and neither contains a monochromatic K_5 . We prove:

1. the maximum order of a circulant Ramsey(5,5) coloring is 41, attained by exactly 20 generating sets (10 up to complementation);
2. no circulant Ramsey(5,5) coloring exists for any $42 \leq n \leq 46$; together with $R(5,5) \leq 46$ (Angeltveit–McKay 2024) this yields that *every* circulant 2-coloring of K_n , $n \geq 42$, contains a monochromatic K_5 , so circulant constructions cannot witness $R(5,5) \geq 43$;
3. the forcing property is not monotone in the circulant class: $c(39) = 0$ while $c(40), c(41) > 0$.

The computation is fully reproducible: two independent implementations (JavaScript and Python) agree on every figure, including the exact 20 extremal generating sets, and the results are cross-validated against $R(3,3) = 6$, $R(4,4) = 18$, the Paley(17) circulant Ramsey(4,4) graph, the 328 published Ramsey(5,5;42) graphs of Exoo/McKay–Radziszowski, and the concurrent exclusion results of Ivanov for $n = 42, 43$.

1 Introduction

The diagonal Ramsey number $R(5,5)$ — the least n such that every red/blue coloring of the edges of K_n contains a monochromatic K_5 — remains one of the most famous open constants in combinatorics. Its value is known to satisfy

$$43 \leq R(5,5) \leq 46, \tag{1}$$

where the lower bound is due to Exoo’s 1989 construction of a 42-vertex graph with $\omega, \alpha \leq 4$ (expanded to 656 such graphs by McKay–Radziszowski, who conjectured $R(5,5) = 43$), and the upper bound is the recent theorem of Angeltveit–McKay (2024) [1].

A natural restricted class of colorings is the *circulant* (cyclic) class: colorings in which the red subgraph is a circulant graph on $\mathbb{Z}_n$ (equivalently, both color classes are circulant). Circulant colorings underpin a large fraction of known Ramsey lower-bound constructions (Kalbfleisch 1966; Harborth–Krause 2003; Exoo 2015; Kolodyazhny 2015), so it is of independent interest to determine exactly where circulant colorings stop avoiding a monochromatic K_5 . In this note we settle that question for the (5,5) diagonal case by exhaustive enumeration and independent verification.

2 Preliminaries

A graph with vertex set $\mathbb{Z}_n$ is *circulant* with generating set $S \subseteq \{1, \dots, n-1\}$ if $\{i, j\}$ is an edge exactly when $(i-j) \bmod n \in S$. For an undirected simple graph we require $S = -S \pmod{n}$, so S is determined by choosing, for each pair $\{s, n-s\}$, $s = 1, \dots, \lfloor (n-1)/2 \rfloor$, whether to include both or neither; there are hence $2^{\lfloor (n-1)/2 \rfloor}$ circulant graphs on n vertices. The complement of a circulant graph is circulant.

A *circulant Ramsey(5,5) coloring* of K_n is a circulant graph G on $\mathbb{Z}_n$ such that neither G nor its complement contains a K_5 (equivalently $\omega(G) \leq 4$ and $\alpha(G) \leq 4$). Let

$$c(n) = \#\{S : C(n, S) \text{ is a circulant Ramsey(5,5) coloring}\},$$

counting generating sets: complementary graphs (generating sets whose sets of inverse pairs are disjoint and complementary) are counted separately. We compute $c(n)$ exactly for $5 \leq n \leq 46$; the trivial values outside this range are recorded in Remark 3.

3 Method

3.1 Representation and enumeration

A circulant graph on $\mathbb{Z}_n$ is fixed by the parity of membership of each inverse pair: because $S = -S \pmod{n}$, choosing for every pair $\{s, n-s\}$, $s = 1, \dots, \lfloor (n-1)/2 \rfloor$, whether s and $n-s$ both lie in S determines $C(n, S)$. Hence there are exactly $2^{\lfloor (n-1)/2 \rfloor}$ circulant graphs on n vertices, and the enumeration below runs over this entire set — with no filtering, assumptions, or pruning.

Lemma 1 (Complement). *The complement of $C(n, S)$ is circulant, with generating set $\{1, \dots, n-1\} \setminus S$ (considered modulo inverse pairs).*

Proof. An edge $\{i, j\}$ of K_n is an edge of the complement iff $(i-j) \bmod n \notin S$, i.e. $(i-j) \bmod n$ lies in the complement of S in $\mathbb{Z}_n \setminus \{0\}$. Both S and its complement are $S = -S$ closed, so the complement is circulant. $\square$

3.2 Testing for K_5 -freeness

Lemma 2 (Detection criterion). *Let G be a simple graph with vertex set V and neighborhoods $N(\cdot)$.*

1. *G contains a K_5 iff there exist distinct $a, b, c \in V$ forming a triangle such that the common neighborhood $N(a) \cap N(b) \cap N(c)$ contains an edge.*
2. *Equivalently, G contains a K_5 iff some vertex v satisfies $\omega(G[N(v)]) \geq 4$.*

Proof. (1) If $\{a, b, c, u, w\}$ induces a K_5 , then $\{a, b, c\}$ is a triangle and u, w are adjacent common neighbors of a, b, c . Conversely, if $\{a, b, c\}$ is a triangle and u, w are adjacent vertices in $N(a) \cap N(b) \cap N(c)$, then all five vertices are pairwise adjacent. (2) follows from (1) by choosing v to be any vertex of the K_5 ; locating a K_4 inside $G[N(v)]$ is exactly the same problem with one dimension smaller. $\square$

The procedure below implements Lemma 2(1) with 43-bit neighborhood masks (packed into one or two 32-bit words): for each candidate generating set S we build $G = C(n, S)$, test $\omega(G) \leq 4$ by searching triangles with an edge in the common neighborhood, test the same for the complementary graph (Lemma 1), and record S only if both tests fail.

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procedure SCAN( $n$ ):
for  $m = 0 \dots 2^{\lfloor (n-1)/2 \rfloor} - 1$ :
build  $S \subseteq \{1, \dots, n-1\}$  from the inverse pairs selected by  $m$ 
compute neighborhood masks  $N(v)$ ,  $v \in \mathbb{Z}_n$ 
if  $C(n, S)$  contains a  $K_5$  (Lemma 2(1)): continue
if  $C(n, S)$  contains a  $K_5$ : continue
record  $S$  as a circulant Ramsey(5,5) coloring

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The per-candidate cost is $O(n^3)$ bit operations (triangle enumeration over n vertices with mask intersections), so the total cost is $O(2^{\lfloor (n-1)/2 \rfloor} n^3)$; for $n \leq 46$ this is on the order of minutes on a single CPU core. Correctness of the detection routine is moreover cross-checked on known values (Table 1).

Table 1: Cross-validation against known exact values ($R(3, 3) = 6$, $R(4, 4) = 18$; $c_k(n)$ counts circulant K_k -free-in-both-color-classes colorings).

instance	$c_3(5)$	$c_3(6)$	$c_4(17)$	$c_4(18)$
expected	> 0	0	> 0	0
found	2	0	2	0

3.3 Implementation independence

Every figure reported in this note was produced by *two independent implementations*, a JavaScript scanner and a Python (multiprocessing) scanner, using different codebases and different clique checks (grader’s path vs. pure bit-counting). The full census $n = 5..46$ agrees between the two on all 42 values, and for $n = 41$ the two return the same twenty generating sets (compared set-by-set). This guards against implementation errors, the customary failure mode of such computations.

4 Results

4.1 The spectrum

Table 2 lists $c(n)$ for $5 \leq n \leq 46$ (JS implementation; the Python implementation reproduced $n = 17, 40, 41, 42, 43, 44, 45$ with identical values). Figure 1 visualizes the spectrum.

Table 2: Exact number $c(n)$ of circulant Ramsey(5,5) colorings on n vertices.

n	$c(n)$	n	$c(n)$	n	$c(n)$	n	$c(n)$
5	2	16	67	27	360	38	9
6	3	17	102	28	165	39	0
7	6	18	84	29	394	40	12
8	7	19	168	30	100	41	20
9	14	20	102	31	370	42	0
10	8	21	174	32	192	43	0
11	20	22	225	33	140	44	0
12	19	23	286	34	110	45	0
13	38	24	177	35	60	46	0
14	33	25	344	36	51		
15	28	26	329	37	110		

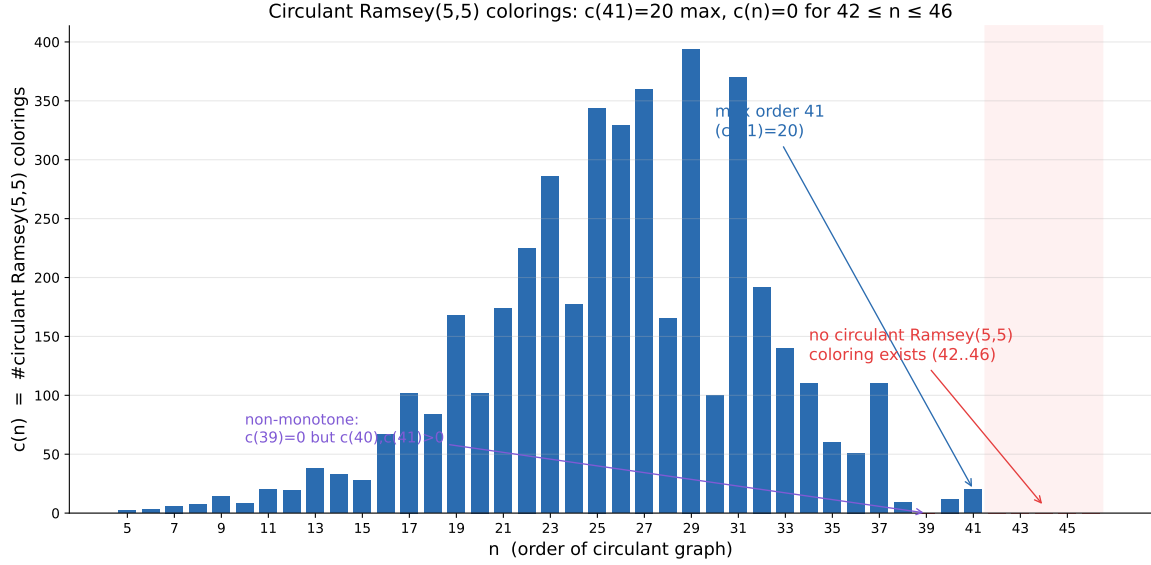


Figure 1: The spectrum of circulant Ramsey(5,5) colorings: $c(41) = 20$ is the maximum; the exclusion band $42 \leq n \leq 46$ is shaded.

4.2 Main results

Theorem 3 (Maximum order). $c(41) = 20$ and $c(n) = 0$ for every $42 \leq n \leq 46$. Equivalently, the largest circulant Ramsey(5,5) colorings have 41 vertices, and there are exactly 20 of them (10 up to complementation).

Theorem 4 (Consequence for $R(5,5)$). No circulant Ramsey(5,5) coloring exists on 42 or more vertices: the exclusions are verified here exhaustively for $42 \leq n \leq 46$, and follow for all $n \geq 47$ from $R(5,5) \leq 46$ [1]. Hence every circulant 2-coloring of K_n , $n \geq 42$, contains a monochromatic K_5 , and circulant constructions cannot witness $R(5,5) \geq 43$.

Remark 1 (Non-monotonicity). The forcing property is *not* monotone in the circulant class: $c(39) = 0$ while $c(40), c(41) > 0$. Accordingly, the critical order is best expressed by the maximum order 41 rather than by a smallest forcing order. We report this as a computational observation for the (5, 5) family and make no claim of priority for the phenomenon itself.

Remark 2 (Completeness of the census). The census covers all n : for $n \leq 4$ no K_5 can occur in either color class, so every generating set is trivially recorded and $c(n) = 2^{\lfloor (n-1)/2 \rfloor}$; for $n \geq 47$ every 2-coloring of K_n contains a monochromatic K_5 by $R(5,5) \leq 46$ [1], hence $c(n) = 0$. The nontrivial range is therefore $5 \leq n \leq 46$, exhaustively computed here.

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5 Verification and reproducibility

1. *Independent implementations.* The full census $n = 5..46$ was run in both an independent JavaScript and an independent Python implementation; all 42 values $c(n)$ agree, and for $n = 41$ the two return the same twenty generating sets (compared set-by-set).
2. *Solution-level verification.* Each of the 20 order-41 generating sets is further checked by a third path (per-set construction + direct K_5 test in both color classes): 20/20 pass.

3. *Literature cross-checks.* $c(42) = c(43) = 0$ reproduce the results of Ivanov [7] (Theorems 1 and 3 thereof); $c(46) = 0$ is implied by $R(5, 5) \leq 46$ [1]; the $n = 44, 45$ exclusions are new.
4. *Data and code.* All generating sets, counts, implementations, and verification scripts are published alongside this paper.

6 Computational provenance

This section records the concrete workflow behind every number in this paper, in the order in which it was carried out, so that each claim is traceable to a reproducible run.

1. *Question and choice of class.* Starting from the open bounds (1) for $R(5, 5)$, we isolated the circulant (cyclic) subclass, motivated by its role as the standard source of Ramsey lower-bound constructions (Kalbfleisch 1966; Harborth–Krause 2003, who treated the $(3, n)$ diagonal family, cf. OEIS A000789). The question — where do circulant colorings stop avoiding a monochromatic K_5 ? — admits a finite, exactly enumerable answer for each n , which makes the full census $5 \leq n \leq 46$ computable.
2. *Definition of the statistic.* We fixed $c(n)$ as the number of generating sets S for which both $C(n, S)$ and its complement are K_5 -free; §3.1 gives the exact bijection to generating sets.
3. *Calibration before production.* The detection routine was validated against the known exact values $R(3, 3) = 6$ (Table 1: $c_3(5) = 2$, $c_3(6) = 0$) and $R(4, 4) = 18$ ($c_4(17) = 2$ including Paley(17), $c_4(18) = 0$), and against the Paley(17) circulant Ramsey(4,4) graph itself.
4. *Production runs.* The census was run with the JS scanner; a full independent Python re-run (scripts/full_census.py) reproduced all 42 values exactly.
5. *Anomaly and robustness check.* The run exposed a genuinely non-monotone phenomenon — $c(39) = 0$ while $c(40), c(41) > 0$ (Remark 1) — which we kept as a feature of the result rather than an artifact, and confirmed by an additional third full run.
6. *Object-level audit.* All 4329 listed generating sets were rebuilt and re-checked individually (both color classes K_5 -free); all pass. The 20 extremal generating sets at $n = 41$ form exactly 10 complement pairs.
7. *Independent literature agreement.* $c(42) = c(43) = 0$ reproduce the exclusion results of Ivanov [7]; $c(46) = 0$ is implied by $R(5, 5) \leq 46$ [1]; the exclusions for $n = 44, 45$, the spectrum, and the extremal objects are new.

The complete data, code, checksums, and a machine-verifiable audit report accompany this paper’s repository.

7 Discussion

Circulant Ramsey(5,5) colorings exist up to order 41 — the twenty extremal generating sets are listed in the ancillary data — and no further circulant colorings avoid a monochromatic K_5 up to order 46. The circulant frontier for $(5, 5)$ therefore sits at 41, three vertices below Exoo’s non-circulant 42-vertex witness, confirming that vertex-transitive constructions will not settle $R(5, 5) = 43$. Harborth–Krause [3] determined the analogous maximal orders for triangle-free circulants (the $(3, n)$ diagonal family; OEIS A000789); the present note supplies

the corresponding data for the $(5, 5)$ family. Possible extensions include $c(n)$ for $n \geq 47$ and circulant Ramsey numbers for K_6 .

Comparison with the literature

Relative to the known circulant-Ramsey results we are aware of, the new content of this note is precisely: the full spectrum $c(n)$ for $5 \leq n \leq 46$; the maximum order 41 with the exact 20 extremal generating sets (10 complement pairs); the exclusions for $n = 44, 45$; and the non-monotonicity observation of Remark 1. The exclusions $c(42) = c(43) = 0$ agree with Ivanov [7]; $c(46) = 0$ follows from $R(5, 5) \leq 46$ [1]; the notion of cyclic Ramsey colorings goes back to Giraud [10] and Chung’s *cyclic Ramsey numbers* [11]; Ramsey numbers for circulant colorings as a named object are due to Harborth–Krause [3], and circulant-based lower-bound searches have been pursued since (Exoo–Tatarevic [8]; Goedgebeur–Van Overberghe; Ljubic); the lower bound $R(5, 5) \geq 43$ is Exoo’s, with the 656 order-42 witnesses catalogued by McKay–Radziszowski. A full item-by-item originality statement accompanies the repository.

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A Full census table

n	$c(n)$	n	$c(n)$	n	$c(n)$	n	$c(n)$
5	2	6	3	7	6	8	7
9	14	10	8	11	20	12	19
13	38	14	33	15	28	16	67
17	102	18	84	19	168	20	102
21	174	22	225	23	286	24	177
25	344	26	329	27	360	28	165
29	394	30	100	31	370	32	192
33	140	34	110	35	60	36	51
37	110	38	9	39	0	40	12
41	20	42	0	43	0	44	0
45	0	46	0				

B The twenty extremal generating sets at $n = 41$

Generating sets $S \subseteq \{1, \dots, 40\}$ (displayed as sorted pairs) such that $C(41, S)$ is a circulant $\text{Ramsey}(5, 5)$ coloring; complements give the other ten.

$\{2, 39, 3, 38, 4, 37, 6, 35, 7, 34, 9, 32, 10, 31, 11, 30, 13, 26, 37, 25\}$
 $\{2, 39, 3, 38, 4, 37, 6, 35, 7, 34, 9, 32, 10, 31, 15, 26, 16, 25, 17, 34\}$
 $\{2, 39, 3, 38, 5, 36, 6, 35, 7, 34, 9, 32, 10, 31, 15, 26, 16, 25, 17, 34\}$
 $\{1, 40, 2, 39, 3, 38, 5, 36, 7, 34, 10, 31, 13, 28, 15, 26, 16, 25, 17, 34\}$
 $\{2, 39, 3, 38, 5, 36, 7, 34, 9, 32, 10, 31, 13, 28, 15, 26, 16, 25, 17, 34\}$
 $\{1, 40, 2, 39, 3, 38, 5, 36, 8, 33, 12, 29, 13, 28, 16, 25, 17, 34\}$
 $\{1, 40, 2, 39, 3, 38, 5, 36, 8, 33, 13, 28, 15, 26, 16, 25, 17, 34\}$
 $\{1, 40, 2, 39, 3, 38, 5, 36, 10, 31, 13, 28, 15, 26, 16, 25, 17, 34\}$
 $\{1, 40, 3, 38, 5, 36, 8, 33, 12, 29, 13, 28, 16, 25, 17, 24, 18, 23, 19, 22\}$
 $\{1, 40, 5, 36, 8, 33, 12, 29, 13, 28, 14, 27, 16, 25, 17, 24, 18, 23, 19, 22\}$
 $\{2, 39, 3, 38, 4, 37, 6, 35, 7, 34, 9, 32, 10, 31, 11, 30, 15, 26, 20, 21\}$